ISyE 6669 FALL 2022 – Practice for Final Exam

Question 1 State University has three professors who each teach four courses per year. Each year, four sections of Marketing, Finance, and Production must be offered. At least one section of each class must be offered during each semester (Fall and Spring). Each professor's time preference and preference for teaching various courses are given in the table below.

	Professor 1	*Professor 2*	*Professor 3*
Fall Preference	3	5	4
Spring Preference	4	3	4
Marketing	6	4	5
Finance	5	6	4
Production	4	5	6

The total satisfaction a professor earns teaching a class is the sum of the semester satisfaction and the course satisfaction. Thus, professor 1 derives a satisfaction of $3+6=9$ from teaching marketing during the fall semester. Formulate a minimum cost network flow problem that can be used to assign professors to courses so as to maximize the total satisfaction of the three professors. (**Hint:** Consider a network having 3 nodes associated with each professor, 3 nodes associated with each course and 6 nodes associated with each pair (semester, course).)

Solution. See the following figure. Each "Professor" node has a net outflow of 4. Each "Course" node has a net outflow of -4. All other nodes have a net putflow of 0. Arcs from, say, Fall-Finance to Finance have a lower bound of 1 to ensure that the minimum number of courses are taught each semester. The cost for each arc from a "Professor" to a "season-course" node is -(reward associated with that teaching assignment). Thus Professor 1 to FallMarketing has a cost of -(3+6)=-9. ■

Question 2 The following network depicts a partially solved minimum cost network flow problem (MCNFP) with labels on nodes indicating net supply and those on arcs showing unit cost, upper bound, and current flow. (The lower bound on the flow through every arc is assumed to be zero.)

(a) (**7 points**) Is the given solution an extreme point (or BFS) for the MCNFP? [**YES:1pt**] If so, what are its basic arcs? [**Thick (1,2),(2,3),(3,5),(4,3),(5,6):2pts**] Specify the nonbasic arcs at lower bound [**Dotted (5,2):2pt**] and the ones at upper bound. [**Solid (1,3)(1,4),(4,6):2pt**]

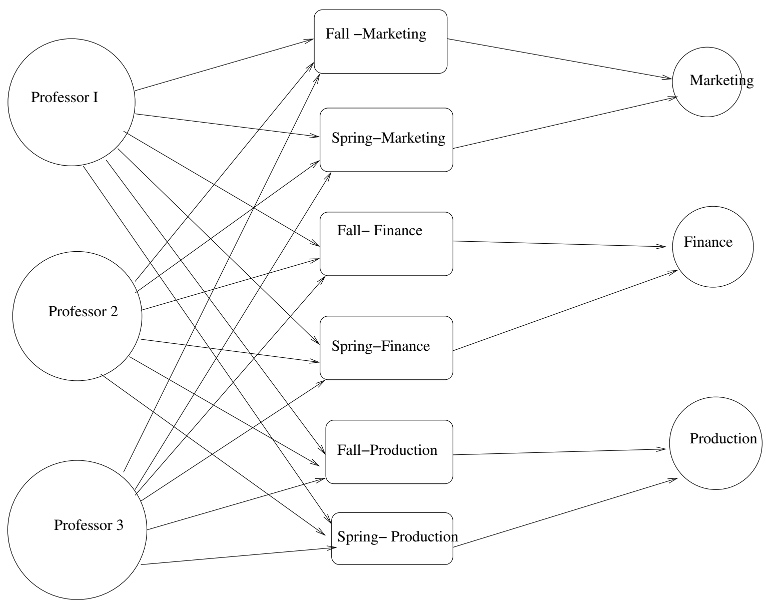


Figure 1: MCNFP formulation

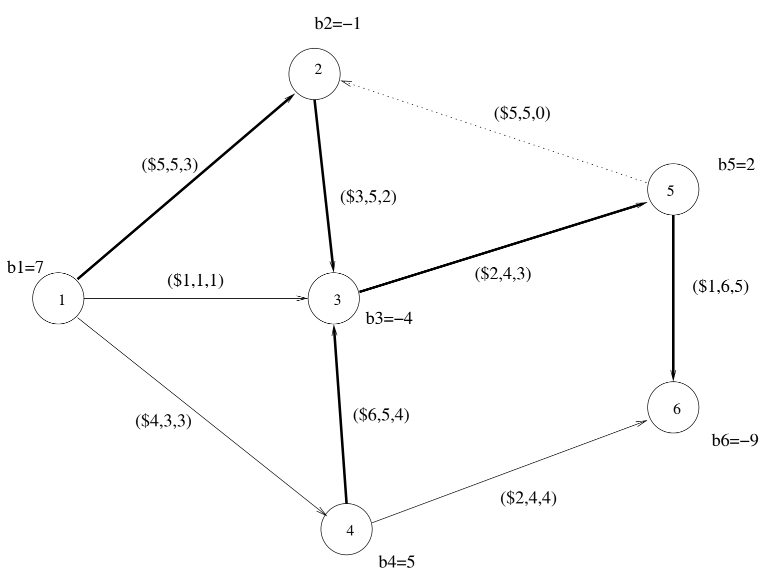


Figure 2: MCNFP Picture

(b) (**7 points**) Consider now performing one iteration of the minimum cost network simplex method. Find the values of the multipliers, that is, the u_i's associated with each node. $[\mathbf{u_1 = 0, u_2 = -5, u_3 = -8, u_4 = -2, u_5 = -10, u_6 = -11}]$

(c) (**8 points**) Determine the reduced costs for each of the nonbasic arcs

$$\mathbf{\bar{c}_{14} = -2, \bar{c}_{13} = 7, \bar{c}_{52} = -10, \bar{c}_{4,6} = 7 \quad (4pts)}$$

and verify that the current solution is not optimal **(3pts):**

$$\begin{aligned}
\mathbf{\bar{c}_{14}} &= -2^{*} \textbf{ negative@upperbound nonbasic, violating optimality condition},\\
\mathbf{\bar{c}_{13}} &= 7 \textbf{ positive@upperbound nonbasic, satisfying optimality condition},\\
\mathbf{\bar{c}_{52}} &= -10 \textbf{ negative@lowerbound nonbasic, satisfying optimality condition},\\
\mathbf{\bar{c}_{4,6}} &= 7 \textbf{ positive@upperbound nonbasic, satisfying optimality condition}
\end{aligned}$$

Which nonbasic arc flow should be modified so that the current flow can be improved? [**(1,4):1pt**]

(d) (**8 points**) Perform one pivot on the "best" available variable. Indicate the entering and leaving variables.[**(1,4) entering, (1,2) leaving: 2pts**]

Give the values of the new BFS obtained. **The variables $\mathbf{x_{14}, x_{43}, x_{23}, x_{12}}$ change to $\mathbf{x_{14} = 1, x_{43} = 2, x_{23} = 4, x_{12} = 5}$ and the other variables remain unchanged. Variable x_{12} becomes nonbasic at upper bound and variable x_{14} becomes basic : 3pts**]

By how much has the flow value improved? [**\$4=reduced cost$\times\theta$: 3pts**]

Question 3 Consider the following LP problem in minimization form:

$$\begin{aligned}
\text{minimize} \quad & w = 6y_1 - 4y_2 + 3y_3 - 6y_4\\
\text{subject to} \quad & y_1 - y_2 + y_3 + y_4 \geq 1\\
& -3y_1 + 2y_2 - y_3 + 2y_4 \leq -2\\
& y_1 \leq 0,\ y_2 \geq 0,\ y_3 \text{ urs},\ y_4 \leq 0.
\end{aligned}$$

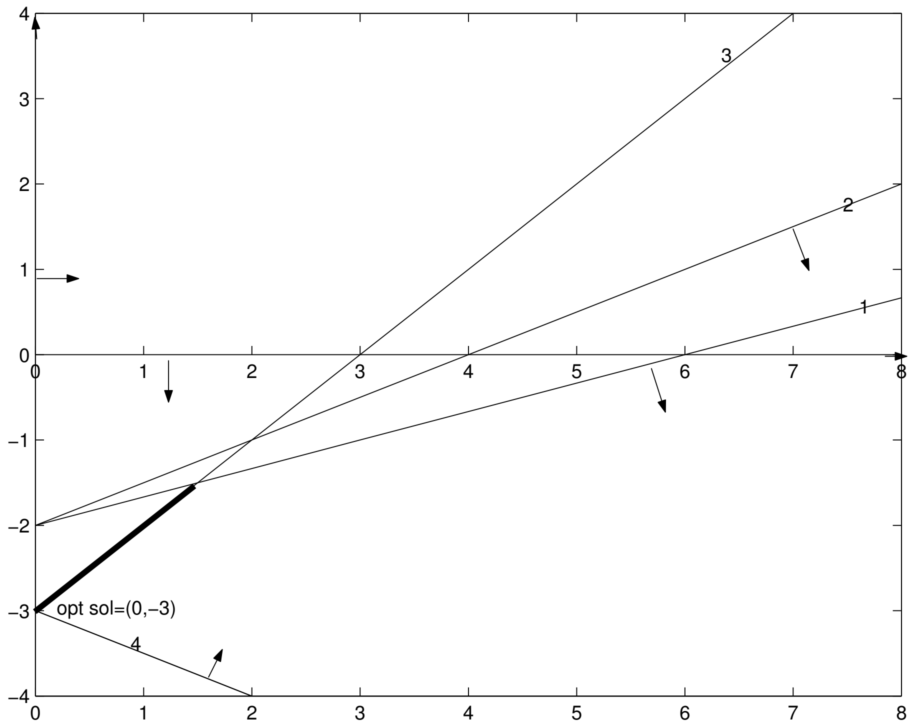
(a) **(9 points)** Find the dual of this problem;

Solution

$$\begin{aligned}
\max \quad & z = x_1 - 2x_2\\
\text{s.t.} \quad & x_1 - 3x_2 \geq 6 && (1)\\
& -x_1 + 2x_2 \leq -4 && (2)\\
& x_1 - x_2 = 3 && (3)\\
& x_1 + 2x_2 \geq -6 && (4)\\
& x_1 \geq 0,\ x_2 \leq 0.
\end{aligned}$$

(b) **(10 points)** Graphically solve the dual problem;

Solution



(c) **(16 points)** Using the complementarity slackness theorem, find the optimal solutions of the above minimization problem. Is the optimal solution unique?

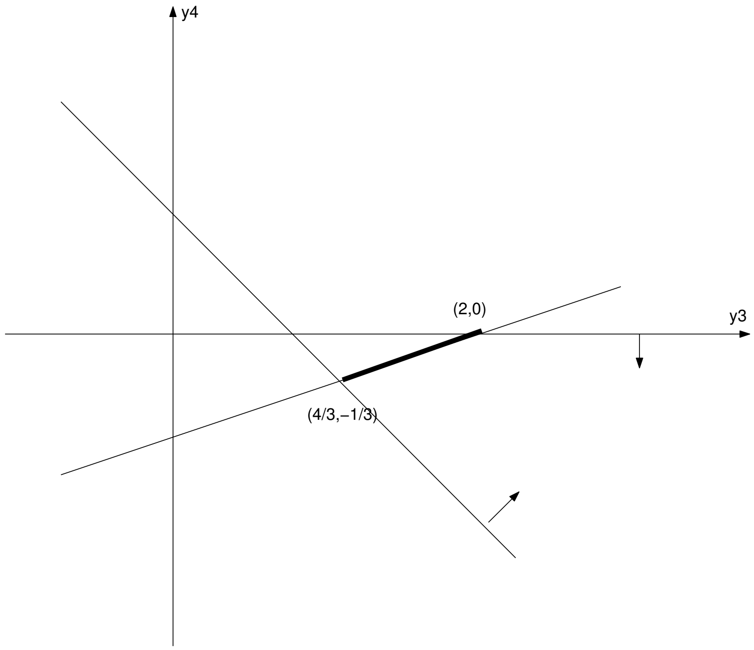
Solution

Complementary slackness condition implies:

$$\begin{array}{ll} x_2 \neq 0 & => -3y_1 + 2y_2 - y_3 + 2y_4 = -2 \\ -x_1 + 2x_2 = -6 < -4 & => y_2 = 0 \\ x_1 - 3x_2 = 9 > 6 & => y_1 = 0. \end{array}$$

Hence, we must have

$y_1 = y_2 = 0$
$-y_3 + 2y_4 = -2$
$y_3 + y_4 \geq 1$
$y_3$urs,$y_4 \leq 0$.



Any point in the segment connecting $(0,0,2,0)$ and $(0,0,\frac{4}{3},-\frac{1}{3})$ is an optimal solution. Hence, the optimal solution is not unique.